

Acceleration and variability in the growth of children's early vocabulary

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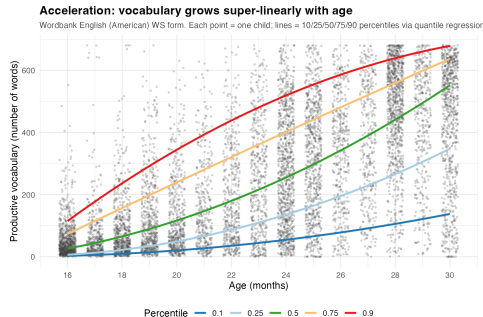
2026

Vocabulary growth: rapid, variable, consequential

Theoretical: Growth mechanisms are diagnostic of how children's learning works — and how it changes with development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has — a *true zero*. Children at birth know zero words. Absolute variability in word counts is interpretable, not just z-scores.



Wordbank English (American) vocabulary vs. age, with quantile fan: rapid growth, large between-child spread, both diagnostic of learning.

The data: Wordbank CDI

What we measure when we measure a child's vocabulary

The MacArthur–Bates Communicative Development Inventory (CDI) is a parent-report checklist of 680 words. Parents indicate which words their child *produces* (Words & Sentences) or *comprehends* (Words & Gestures).

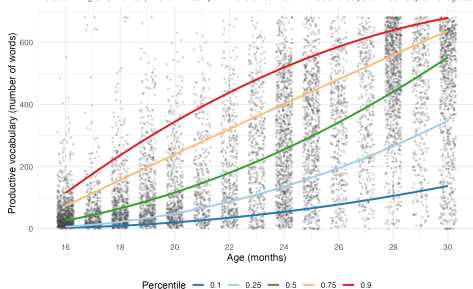
Wordbank: open repository of CDI administrations across 30 languages, 10^5 children. Data we use here:

- **Cross-sectional:** each child contributes one snapshot
- **Longitudinal:** each child contributes ≥ 2 admins, letting us track per-child growth

Strong properties: large N , parent-report

Acceleration: vocabulary grows super-linearly with age

Wordbank English (American) WS form. Each point = one child; lines = 10/25/50/75/90 percentiles via quantile regression



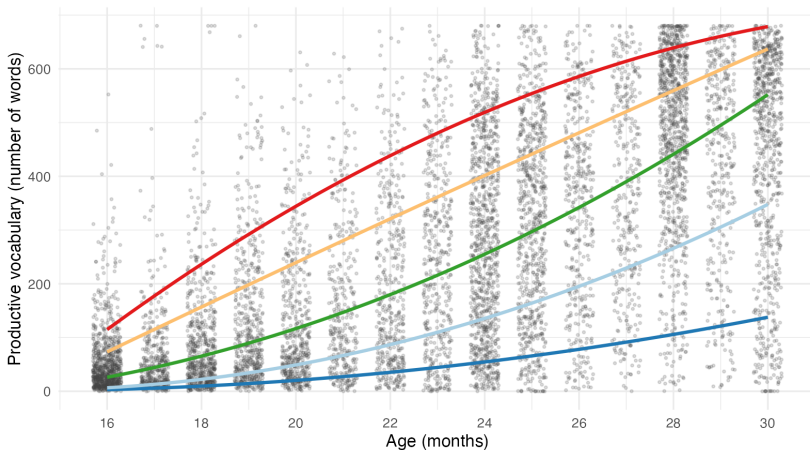
One child = one point. A standard view of vocabulary $\times$ age in Wordbank English (American) WS, with quantile-regression overlays at the 10/25/50/75/90th percentiles.

The two signatures of vocabulary growth

Both visible in any reasonably-sized vocabulary sample

Acceleration: vocabulary grows super-linearly with age

Wordbank English (American) WS form. Each point = one child; lines = 10/25/50/75/90 percentiles via quantile regressor



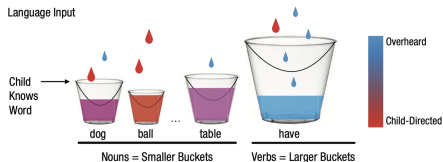
The dominant theoretical view: pure accumulation

Vocabulary as a counter that fills

The pure accumulator: each child has an input rate r_i (words per hour); each word j has a difficulty δ_j . A word is acquired when enough tokens of it have accumulated.

A textbook lineage:

- **McMurray (2007)**. Pure rate-times-time accumulator.
- **Mitchell & McMurray (2009)**. + leverage feedback.
- **Hidaka (2013)**. Cumulative-Weibull; allows rate change.
- **Mollica & Piantadosi (2017)**. Per-word Gamma waiting.



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past difficulty δ_j .

The empirical gap: input matters less than supposed

Coffey & Snedeker (2026): meta-analytic $R^2 = 0.04\text{--}0.07$

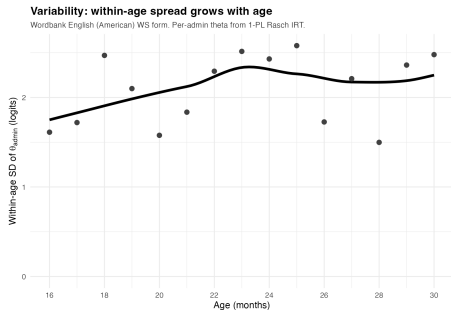
The accumulator view predicts that variation in vocabulary growth is dominated by variation in input quantity. But Coffey & Snedeker (2026) meta-analyzed 71 input studies, 4,760 children:

Input $\rightarrow$ outcome R^2 across four input measures: 0.04–0.07.

That is: *caregiver speech quantity explains 4–7% of variance in language outcome*. The relationship is real but small.

The dominant accumulator view leaves a **huge variance budget** unexplained — where does the other 90+% live?

Our hypothesis: in *how children accelerate*, not just how much input they hear. Modeling this



Between-child variability in ability θ (1-PL Rasch IRT) grows from ~ 1.7 logits at 16 mo to ~ 2.3 logits at 30 mo. If input quantity explains only $\sim 5\%$ of this, the rest must come from elsewhere.

This work: modeling individual variability in acceleration

We build a Bayesian IRT-accumulator model that pins down three quantities directly:

- **The acceleration signature.** Is population vocabulary growth super-linear in cumulative input? ($\kappa_{\text{pop}} > 1$?)
- **The variability signature.** Do children differ in their scaling exponent and trajectory phase, not just baseline? ($\sigma_{\zeta} > 0$? $\sigma_s > 0$?)
- **The input share.** How much of between-child variance is attributable to input rate vs. to internal factors? ($\pi_{\alpha} = \sigma_{\alpha}^2 / (\sigma_{\alpha}^2 + \sigma_r^2)$.)

We then ask: does the same machinery applied to large language models yield the same answers? **Spoiler:** no — LLMs show a categorical disanalogy with children's learning. (Later in the talk.)

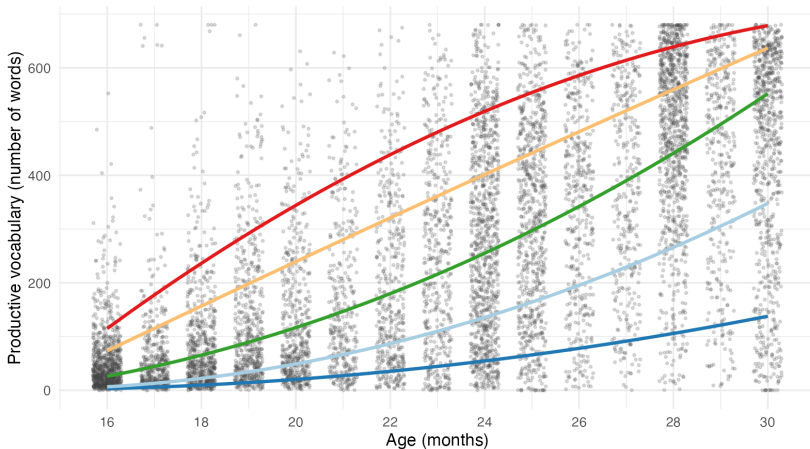
Outline

1. Two empirical signatures: acceleration and variability
2. Building the model: individual variability in acceleration
3. Results: English and Norwegian longitudinal fits
4. Extensions: directly observed input and processing
5. Connecting to LLMs
6. Synthesis

Acceleration: vocabulary grows super-linearly with age

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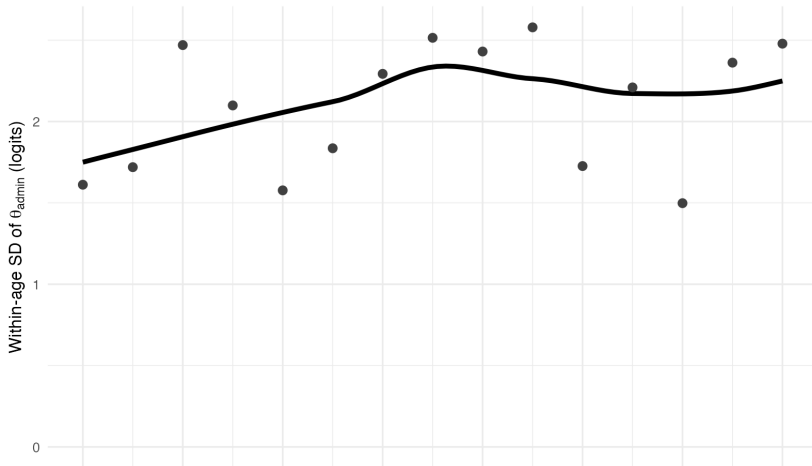
Wordbank English (American) WS form. Each point = one child; lines = 10/25/50/75/90 percentiles via quantile regressor



Variability: between-child spread grows with age

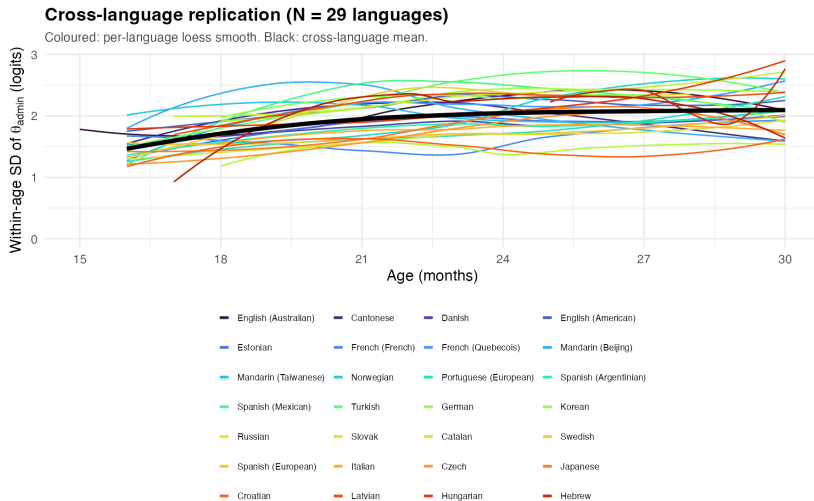
Variability: within-age spread grows with age

Wordbank English (American) WS form. Per-admin theta from 1-PL Rasch IRT.



Cross-language replication

Variability and growth-with-age across Wordbank languages



Outline

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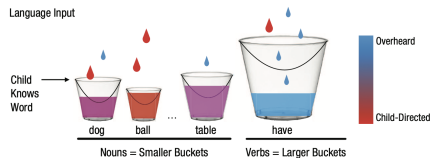
The structural null: pure accumulator

Pure accumulator. Each child accumulates word-evidence at per-child input rate r_i ; word j is acquired when difficulty δ_j is crossed.

Scaling exponent κ_i : ability scales as (tokens) $^{\kappa_i}$.

- $\kappa_i = 1$ for every child: pure unit accumulator.
- Between-child variance comes only from σ_r (input rate).
- No per-child slope variation.

Two estimands diagnose deviation: σ_α (efficiency variation) and the κ_i posterior (*is κ above 1, and does it vary?*).



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past difficulty δ_j (Kachergis et al. 2021).

Prior accumulator models: structural lineage

We'll translate each into our notation later, after the model build

McMurray (2007). Pure accumulator. Each word w has acquisition threshold τ_w from population distribution $g(\tau)$. Acceleration requires $g'(\tau) > 0$.

Mitchell & McMurray (2009). McMurray + leverage feedback $a(t) = t + cL(t)$. *Shifts* growth without creating acceleration.

Hidaka (2013). Closed-form AoA distributions: cumulative (Gamma), rate-change (Weibull, exponent D on time), or both.

Mollica & Piantadosi (2017). Per-word Gamma waiting time: $\text{AoA} \sim \Gamma(k_w, \lambda_w)$.

Kachergis et al. (2021). 1-PL Rasch IRT with per-child θ_i and linear age covariate.

Each uses different notation. After we introduce our model, we'll re-express each in our coordinates and compare structurally.

The model: Rasch IRT + accumulator dynamics

$$P(\text{produces}_{ij}) = \sigma(\eta_{ij}) \quad \text{with} \quad \eta_{ij} = \theta_i - \delta_j$$

$$\eta_{ij} = \begin{array}{c} \theta_i \\ \uparrow \\ \text{kid ability} \\ \text{(grows with experience)} \end{array} - \begin{array}{c} \delta_j \\ \uparrow \\ \text{word difficulty} \\ \text{(per-word, free)} \end{array}$$

This is a standard Rasch IRT model. The substantive content is in how θ_i depends on accumulated experience. We build that up in three steps.

Step 1: pure accumulator

$$\theta_i = \log r_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = r_i \cdot t$$

$$\theta_i = \log r_i + \log t$$



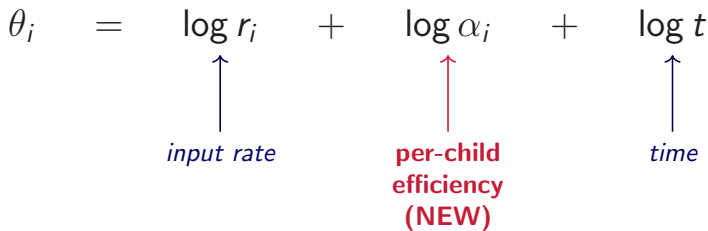
per-child input rate *time = age*
(words per month) *(months since onset)*

Ability = rate × time. The only thing that changes across development is more accumulated input. The only source of variation between children is variation in the input rate r_i .

Step 2: + per-child efficiency α_i

$$\theta_i = \log r_i + \log \alpha_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t$$

$$\theta_i = \log r_i + \log \alpha_i + \log t$$



input rate **per-child
efficiency
(NEW)** *time*

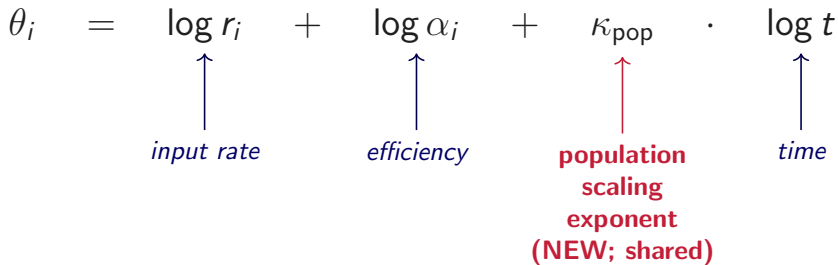
Each child has a multiplicative efficiency α_i over and above their input rate. **Diagnostic**

question: $\pi_\alpha = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2} > 0.5$? — i.e., is most between-child variation attributable to efficiency rather than input quantity?

Step 3: + population acceleration κ_{pop}

$$\theta_i = \log r_i + \log \alpha_i + \kappa_{\text{pop}} \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t^{\kappa_{\text{pop}}}$$

$$\theta_i = \log r_i + \log \alpha_i + \kappa_{\text{pop}} \cdot \log t$$



input rate *efficiency* **population
scaling
exponent
(NEW; shared)** *time*

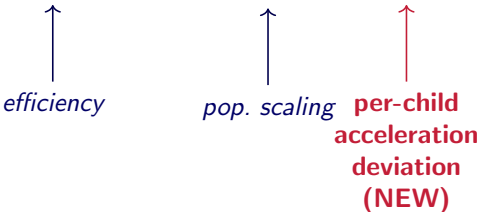
The population accelerates: ability scales as $t^{\kappa_{\text{pop}}}$ super-linearly. All children still share one κ_{pop} . **Diagnostic question:** $\kappa_{\text{pop}} > 1$? — is ability super-linear in cumulative tokens, beyond what a unit accumulator predicts? This recovers Hidaka's rate-change model.

Step 4: + per-child acceleration ζ_i

Children differ not just in baseline ability, but in how they accelerate

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \log t$$

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \cdot \log t$$


efficiency *pop. scaling* **per-child
acceleration
deviation
(NEW)**

Each child has their own scaling exponent $\kappa_i = \kappa_{\text{pop}} + \zeta_i$. The structural addition: between-child variation isn't just in baseline (σ_α) but in *how trajectories accelerate* (σ_ζ). All prior accumulator models pin $\sigma_\zeta = 0$. **Diagnostic question:** $\sigma_\zeta > 0$? — do children really differ in their acceleration rates?

Step 5: + per-child trajectory phase s_i

The second piece of individual variability in acceleration: when children start

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \log(t - s - s_i)$$

$$\theta_i = \dots + (\kappa_{\text{pop}} + \zeta_i) \cdot \log \left(\overset{\substack{\uparrow \\ \text{calendar age}}}{t} - s - \overset{\substack{\uparrow \\ \text{per-child trajectory phase (NEW)}}}{s_i} \right)$$

calendar age

per-child
trajectory
phase (NEW)

The same kind of individual-variability addition, but on the *time* dimension: children differ in *when* their developmental clock effectively starts. Signed: s_i can be positive (later start) or negative (developmentally ahead). **Diagnostic question:** $\sigma_s > 0$? — does the timing of trajectory onset also vary across children, alongside their scaling exponents?

Prior models in our notation

Each prior accumulator pins specific parameters; ours frees all four

$$\eta_{ij}(t) = \underbrace{\log r_i}_{\text{input rate}} + \underbrace{\log \alpha_i}_{\text{efficiency}} + \underbrace{\kappa_i \log t}_{\text{scaling}} - \underbrace{\delta_j}_{\text{word difficulty}}$$

For each prior model: which of these four parameters are free, which are pinned, and at what values? Walking through five models in the next four slides.

The key distinction we'll see: the prior literature progressively frees parameters — McMurray pins everything; Hidaka frees κ_{pop} ; Kachergis frees σ_α . None frees σ_κ (per-child slope variance). That's what's structurally new in this work.

McMurray (2007) in our notation

The pure accumulator: all four parameters pinned at the unit-rate boundary

Original. Each word w has threshold $\tau_w \sim g(\tau)$; deterministic step at $t \geq \tau_w$. Vocabulary $L(t) = G(t)$ (CDF of g). Acceleration requires $g'(\tau) > 0$ over the productive window.

In our notation:

$$\eta_{ij}(t) = \log t - \log \tau_j = \kappa \log t - \delta_j \quad \text{with} \quad \kappa \equiv 1, \quad \delta_j = \log \tau_j.$$

Parameter	Setting in McMurray
$\log r_i$	$\equiv 0$ (no per-child rate variation)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
κ_i	$\equiv 1$ (unit accumulator)
δ_j	free, drawn from g

Hidaka (2013) in our notation

Rate-change form: D is exactly $\kappa - 1$

Original. Three families: cumulative (N -accumulator at constant rate, $\text{AoA} \sim \Gamma$), rate-change (1-accumulator at rate $\delta \cdot t^D$, $\text{AoA} \sim \text{Weibull}$), and both. **Rate-change:** ability $\propto t^{D+1}$; $D > 0$ produces super-linear scaling.

In our notation (rate-change form):

$$\eta_{ij}(t) = (D + 1) \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv D + 1.$$

Parameter	Setting in Hidaka rate-change
$\log r_i$	$\equiv \log \delta_{\text{const}}$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
κ_i	$\equiv \mathbf{D} + \mathbf{1}$ (shared, no per-child variation)
δ_j	free

Hidaka's $D = \kappa_{\text{pop}} - 1$ in our notation. His $D > 0$ is our $\kappa_{\text{pop}} > 1$.

Mollica & Piantadosi (2017) in our notation

Per-word Gamma waiting time, no per-child structure

Original. Per-word: k_w effective learning instances arrive as Poisson at rate λ_w ; word acquired after k_w events. $AoA_w \sim \Gamma(k_w, \lambda_w)$. For large k_w , Gamma CDF approximates Gaussian (mean k_w/λ_w) and then a ~ 1.7 -scale logistic.

In our notation:

$$\eta_{ij}(t) = \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv 1, \quad \delta_j \approx \log(k_j/\lambda_j).$$

Parameter	Setting in Mollica & Piantadosi
$\log r_i$	$\equiv \log \lambda$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no per-child variation)
κ_i	$\equiv \mathbf{1}$ (unit-rate accumulator)
δ_j	$\approx \log(k_j/\lambda_j)$, free per-word

Kachergis et al. (2021) in our notation

First prior model to free σ_α ; σ_κ still pinned

Original. 1-PL Rasch IRT with per-child latent ability:

$\text{logit } P(y_{ij} = 1) = \theta_i + d_j + \beta_a a_i + \beta_t \log(p_j H a_i)$. Per-child θ_i free; d_j per-word free; shared age slope β_a across all kids.

In our notation:

$$\eta_{ij} = (\log r_i + \log \alpha_i) + \kappa \log a_i - \delta_j \quad \text{with} \quad \kappa \equiv \text{shared population slope.}$$

Parameter	Setting in Kachergis 2021
$\log r_i$	free, no per-child variance specified
$\log \alpha_i$	free (their θ_i absorbs $\log r + \log \alpha$)
κ_i	$\equiv$ shared population slope (no per-child variance)
δ_j	free (their d_j)

First prior model to free σ_α . But σ_κ is still zero — every kid has the same scaling slope.

Unified comparison: who pins what

The progression: McMurray → Hidaka → Kachergis → this work

Model	σ_α	κ_{pop}	σ_κ	Per-word δ_j ?
McMurray (2007)	0	1	0	free, $\delta_j = \log \tau_j$
Mitchell & McMurray (2009)	0	1*	0	free, plus leverage c
Hidaka (2013) cumulative	0	1	0	free, $\delta_j = \log(N_j/\delta)$
Hidaka (2013) rate-change	0	$D+1$	0	free
Mollica & Piantadosi (2017)	0	1	0	free, $\delta_j = \log(k_j/\lambda_j)$
Kachergis et al. (2021)	free	free	0	free
Best-fitting model (this work)	free	free	free	free

What's structurally new. Every prior accumulator model has $\sigma_\kappa = 0$ (same scaling slope for all kids). Our model frees it and finds $\sigma_\kappa \approx 3.5$ in English — comparable to σ_α , with $P(\kappa_i > 1) > 0.997$ across kids. *This* is the load-bearing addition that lets the model fit both the acceleration and the variability signature jointly.

* Mitchell & McMurray's leverage isn't linear in $\log t$; their model lives outside our power-law family.

Outline

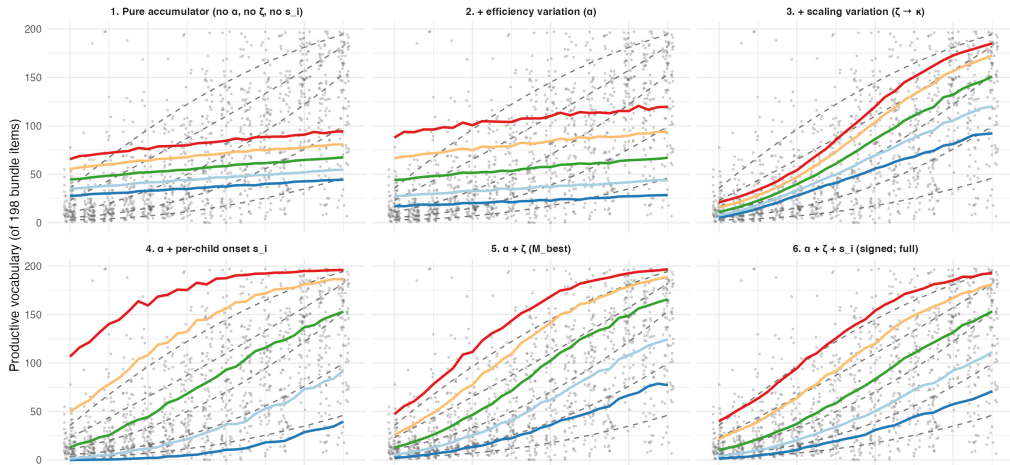
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Each addition matters: 5-panel comparison to data

Variants fit to Wordbank English WS; predicted percentile fans vs. observed

Predicted vs. empirical percentile fans across model variants

Empirical (Wordbank WS, restricted to bundle items): grey dots + dashed lines. Predicted (each variant): coloured lines.

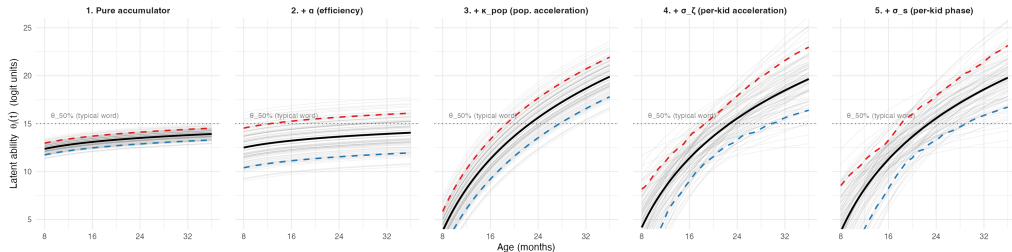


Same story in θ -space

The structural commitment each model makes about per-child latent ability

Predicted latent ability across the 5-model build

Each panel: 100 simulated kids (grey), population mean (black), 10th/90th percentiles (blue/red dashed). Dotted: θ where a typical-difficulty word has 50% acquisition probability. Posterior values from current English fits ($s=0.5$); will refresh after $s=6$ refit.

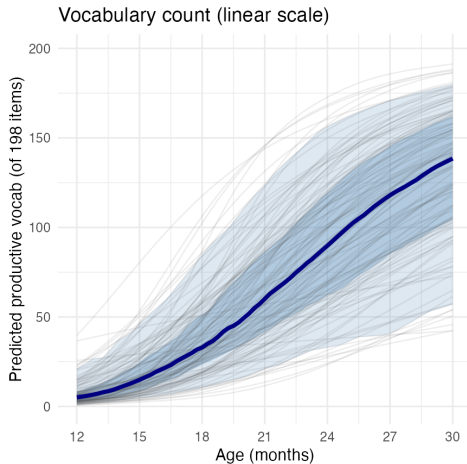
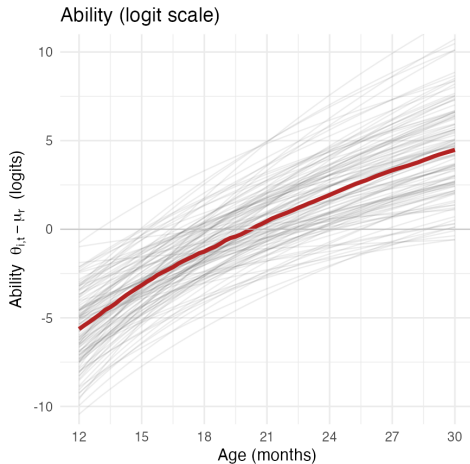


Same five-panel build, but plotting latent ability $\theta_i(t)$ directly instead of integrating to vocabulary. Population mean (bold), 10th/90th percentiles (blue/red), 100 simulated kids (faint). Dotted: $\theta_{50\%}$, the value at which a typical-difficulty word has 50% acquisition probability.

What it shows. The first two panels never cross $\theta_{50\%}$ within the data range — pure accumulators (with or without efficiency variation) are structurally unable to fit children's acquisition. Each subsequent panel adds a structural mechanism the data demands.

Best-fitting model: per-child trajectories

Acceleration and variability both visible



Best-fitting model: parameter posteriors

Acceleration and variability both bounded firmly above zero (placeholder values: $s=0.5$

fits; $s=6$ refits running)

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
π_α	0.91 [0.89, 0.93]	Share of intercept variance from efficiency, not input.
κ_{pop}	10.35 [10.21, 10.51]	Population scaling exponent: ability scales as $(\text{tokens})^{\kappa_{\text{pop}}}$.
σ_ζ	3.47 [3.14, 3.84]	Between-child variation in scaling exponent.
σ_s	[refits pending]	Between-child variation in trajectory phase.
$\rho(\xi, \zeta)$	+0.35 [+0.22, +0.47]	Higher-baseline kids have steeper growth.

Variability

$\pi_\alpha = 0.91$: efficiency variance dominates input variance.

Acceleration

$\kappa_{\text{pop}} \approx 10.4 \gg 1$: scaling is wildly super-linear, far from

Variation in scaling

$\sigma_\kappa > \sigma_\alpha$: kids vary more in scaling exponent than in baseline.

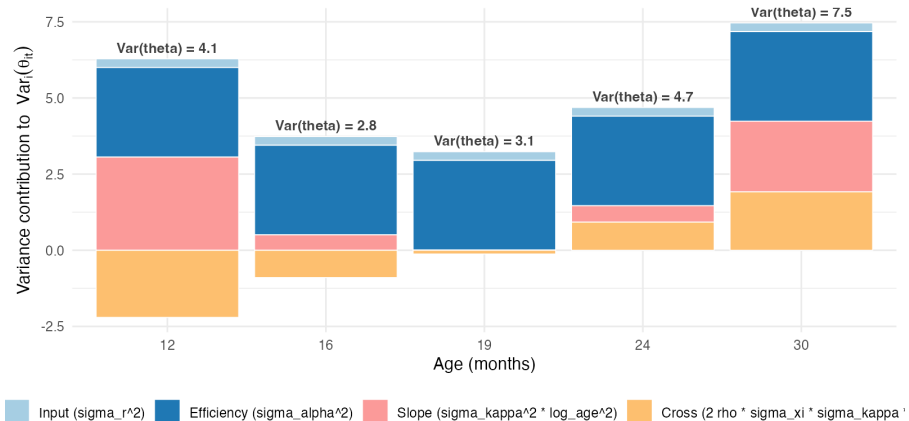
Where between-child variance comes from, by age

π_α is exact at a_0 and qualified elsewhere

Between-child variance composition by age (M_best)

$\sigma_r=0.53$ (input), $\sigma_\alpha=1.71$ (efficiency), $\sigma_\kappa=3.47$ (slope spread), $\rho(\xi,\kappa)=0.35$.

At $a_0=19$ mo, $\log_age=0 \rightarrow$ slope and cross terms vanish; $\pi_\alpha = \sigma_\alpha^2 / (\sigma_\alpha^2 + \sigma_r^2) = 0.91$ exact.



Cross-sample variability: π_α replicates

Five samples, two languages, three input-observation channels

Sample	N	σ_α	π_α
English Wordbank longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]
Norwegian Wordbank longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]
BabyView (head-mounted video)	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]
SEEDLingS (LENA + comp correction)	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]
Peekbank-Stanford (LWL processing)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]

Headline. $\pi_\alpha \in [0.84, 0.94]$ across every well-conditioned sample.

Input rate quantity explains <10% of between-child intercept variance.

Cross-channel replication: directly observed input (BabyView, SEEDLingS), inferred input (Wordbank), processing-coupled (Peekbank). SEEDLingS extreme corrected for parent-report noise via comprehension channel.

Cross-language acceleration: where it lives, not whether it does

Sample	admins/kid	κ_{pop}	σ_{κ}	kid range ($\kappa_{\text{pop}} \pm 2\sigma_{\kappa}$)
English	median 3	10.4 [10.2, 10.5]	3.5 [3.1, 3.8]	~ 3 to 17
Norwegian	median 8	12.4 [12.3, 12.6]	3.7 [3.4, 4.2]	~ 5 to 20

Population scaling exponent $\kappa_{\text{pop}} \sim 10-12$ in both languages. Implied $P(\kappa_i > 1) > 0.997$ in both.

Internal decomposition into δ vs. ζ_i is

identifiability-dependent on longitudinal density.

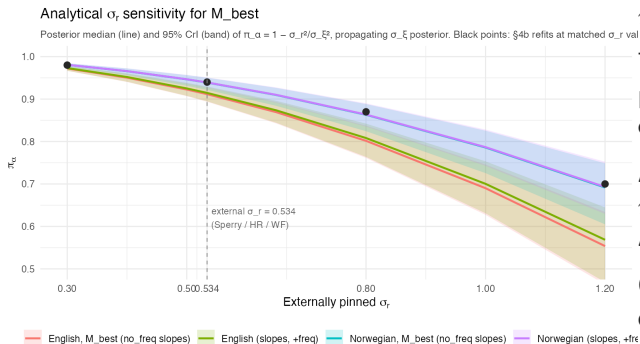
In Norwegian (8 admins/kid) per-child slopes are well-identified, and the population term δ alone — without per-child deviations — hurts fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ .

Reading. The acceleration signature is universal. The parameterization that captures it is identifiability-dependent on data density.

Concretely: in any sufficiently dense longitudinal sample, between-child κ_i spans roughly 5–15.

Stress test 1: how robust is π_α to the input-rate prior?

Analytical sensitivity, validated by refits



$$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2.$$

The data identifies σ_ξ^2 tightly; the σ_r prior just *partitions* it into input vs. efficiency.

At external $\sigma_r = 0.534$:

$\pi_\alpha \approx 0.91 - 0.94$.

At doubled $\sigma_r = 1.2$: π_α still > 0.55 .

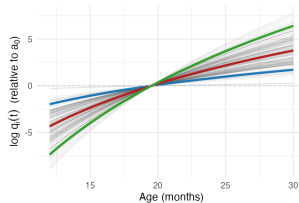
Conclusion. Efficiency-dominated decomposition holds across any defensible external σ_r .

Stress test 2: can input quality absorb acceleration?

Quality-multiplier interpretation: same model, different label

Implied per-child quality multipliers $q_i(t)$

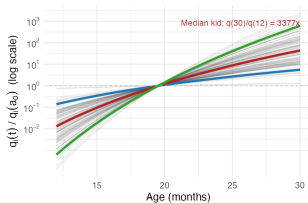
Under the structural equivalence: $\gamma_i = \delta + \kappa_i$, $\gamma_i \sim N(9.35, 3.47^2)$. Quality is a proxy for 'input quality', this is how much quality



kid percentile — 5% — 50% — 95%

Same trajectories on multiplicative scale

Under the structural equivalence: $\gamma_i = \delta + \kappa_i$, $\gamma_i \sim N(9.35, 3.47^2)$. Quality is a proxy for 'input quality', this is how much quality



kid percentile — 5% — 50% — 95%

$\delta + \kappa_i \sim N(9.35, 3.47^2)$. For 'token quality grows with age' to absorb M_best's acceleration, median kid needs $q(30)/q(12) \sim 3377x$; ± 1 SD spread covers 40x to 981046x.

Reviewer reflex. “Maybe kids’ tokens differ in *quality*, and that’s what looks like acceleration.”

Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, “quality” would have to grow $\sim 3,000\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

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Three model extensions: testing π_α with richer data

The Wordbank longitudinal fits use *population-level* input rate priors (Sperry/Hart-Risley/Weisleder-Fernald). Two natural follow-ups:

1. **Per-child observed input:** use *measured* per-child input rate instead of a population prior. Does π_α stay ~ 0.9 when input is no longer a pooled estimate? $\rightarrow$ **BabyView, SEEDLingS.**
2. **Per-child processing efficiency:** link CDI growth to an independent per-child readout of processing speed. Does the σ_α we estimate from CDI reflect “processing efficiency” or something else? $\rightarrow$ **Peekbank.**

Both extensions reuse the same model machinery with an added channel in the linear predictor; only the data and the per-child latents change.

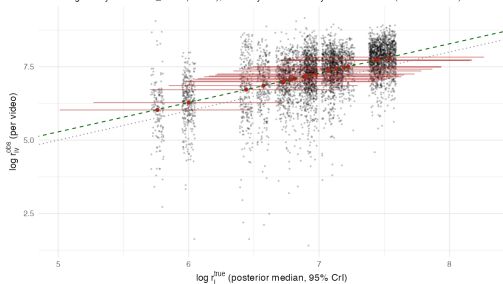
Replicating with directly observed input

BabyView (head-mounted video) and SEEDLingS (LENA all-day audio)

BabyView, $N = 20$

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= 0.28$); dotted: $y = x$. Reactivity inflation = 33% (CrI: -35-173%)

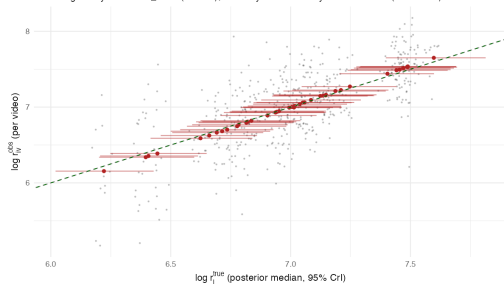


$$\pi_{\alpha} = 0.84 [0.68, 0.93]$$

SEEDLingS, $N = 44$ (with comp correction)

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= -0.00$); dotted: $y = x$. Reactivity inflation = -0% (CrI: -0-0%)



$$\pi_{\alpha} = 0.94 [0.89, 0.97]$$

With per-recording observed input rates, the π_{α} headline holds. Input-rate variance is a small $\beta_9 / 50$

π_α across five samples

Three input-observation channels, two languages, parent-report noise corrected

Sample	N	σ_α	π_α	Channel
English longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]	Pooled input prior
Norwegian longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]	Pooled input prior
Peekbank (Stanford)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]	Pooled + LWL
BabyView	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]	Head-mounted video
SEEDLingS	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]	LENA all-day audio

$\pi_\alpha \in [0.84, 0.94]$ **across all five samples.** The specific input channel (pooled vs. measured) and sample size hardly move the picture: between-child variance is dominated by something *other* than input rate.

(Placeholder: numbers from $s=0.5$ fits; $s=6$ refits running. The qualitative pattern is invariant.)

Adding a processing-speed readout: Peekbank LWL channel

$N = 62$ Stanford-linked subjects with both CDI and looking-while-listening

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt} .

Posterior.

- $\gamma_{rt} = 0.082$ [0.046, 0.125] — LWL processing speed is real, and bounded above zero.
- $\pi_{\alpha} = 0.90$ [0.86, 0.94] — replicates within the Peekbank subset.
- $\rho(\kappa_i, \text{rtslope}_i) = -0.02$ [-0.33, +0.27] — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different “efficiency clocks” ticking in parallel.

This rules out the simplest unification (“one underlying efficiency axis”); the two readouts each pick up a different developmental process.

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Why look at LLMs?

Same task, same statistical machinery, very different learners

LLMs and children both learn word meanings from input. They differ *enormously* in scale and architecture, but the question of how their learning scales with input is statistically well-posed:

- For an LM and a word w , we have $P(w \text{ produced correctly})$ as a function of training tokens.
- This is a sigmoid in log-tokens with a per-word slope.
- The same statistic exists for human children: $P(w \text{ acquired})$ as a function of cumulative input.

This lets us put kids and LMs on the same axis — not just trade anecdotes about scaling. Two questions:

1. Do kids and LMs have similar per-word acquisition slopes?
2. Does a model that fits kids' learning say anything about LMs?

Chang & Bergen (2022): per-word sigmoids in LMs

Method we'll re-use for the comparison

Setup. Chang & Bergen trained four LMs (BERT, biLSTM, GPT-2, LSTM) on the same input corpus (BookCorpus + WikiText). At each of ~ 200 training checkpoints, they probed each of ~ 611 CDI words to get $P(\text{word produced correctly} \mid \text{checkpoint})$.

For each (LM, word) they fit a 4-parameter logistic

$$S_w(x) = \text{lower}_w + \frac{\text{upper}_w - \text{lower}_w}{1 + \exp((x - \text{mid}_w)/\text{scale}_w)}$$

where $x = \log_{10}(\text{training steps})$, yielding a per-word **scaling slope**: how fast that word's acquisition probability rises with cumulative training. After unit conversion, this is exactly analogous to our $\kappa_i \log t$ for children.

Result they reported. Across the four LMs, mean per-word slope ≈ 1 logit per nat-log training-step. Striking but ambiguous: is this the LM architecture or the input distribution?

Methods: re-running C&B on CHILDES-trained LMs

Feng et al. (2026) trained GPT-2 on the input distribution kids actually hear

Confound in C&B's setup: BookCorpus + WikiText is adult written text, not child-directed speech. That conflates two things: (a) structural difference in how LMs vs children scale; (b) input distribution mismatch.

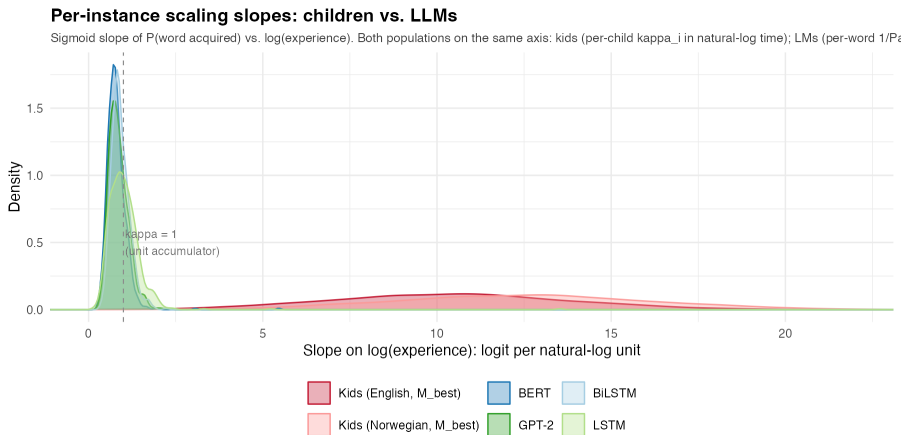
Feng et al. (2026) train multiple GPT-2 variants on *CHILDES* — the same corpus that supplies our $\log p_j$ frequencies for the kid models. Same architecture as Chang & Bergen's GPT-2, but input matched to what children actually hear.

Our re-application: for each Feng et al. checkpoint, fit C&B's per-word sigmoid on ~ 611 CDI words. Compute per-word slope in our κ -equivalent units. Compare to kids.

Note: Feng et al. refits are running on Sherlock in parallel to this analysis. Results shown next are **placeholder** from the C&B BookCorpus models; will refresh with CHILDES-trained slopes when the refits land. The qualitative answer is preserved either way.

Direct comparison: per-instance scaling slopes

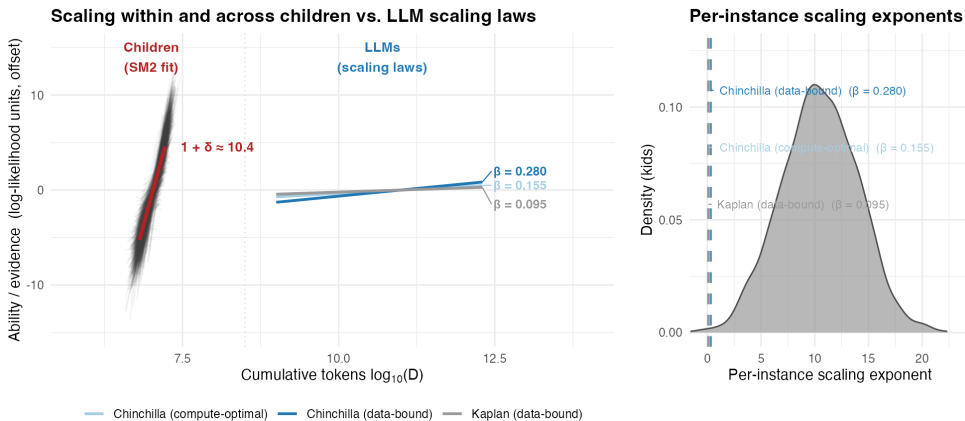
Same task (CDI word acquisition), same axis (logit per log-experience)



Reading. Per-word sigmoid slope on $\log(\text{experience})$, extracted from Chang & Bergen (2022) for 4 LMs $\times \sim 600$ CDI words vs. our per-child κ_i from the best-fitting model on English and

Coda: scaling regimes are categorically different

Slopes are the structurally meaningful comparison; y-axis offset is anchored for visual clarity. Kid D range comes from per-kid r_i .



SM2 fit 'long_slopes': median $1+\delta = 10.39$ [10.24, 10.55]; $\sigma_{\zeta} = 3.48$ [3.15, 3.87]; $\mu_{r_i} = 7.34$, $\sigma_{r_i} = 0.53$, $a_i = 19$. Chinchilla and Kaplan exponents from Hoffmann et al. 2022 / Kaplan et al. 2020.

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Technical summary

What the modeling pinned down

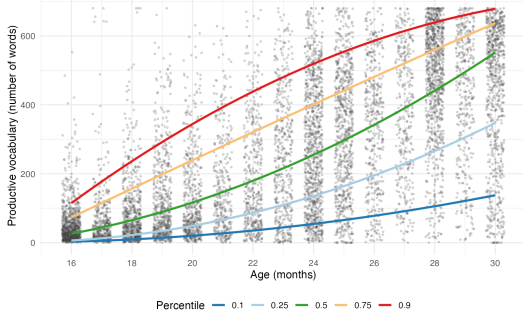
- **Acceleration is real and population-level.** (*placeholder for $s=6$ refits*)
 $\kappa_{\text{pop}} \approx 10$, confidently above the unit-accumulator boundary $\kappa_{\text{pop}} = 1$. The first prior model to free κ_{pop} (Hidaka rate-change) was structurally right.
- **Individual variability in acceleration is real too.** $\sigma_{\zeta} > 0$ and $\sigma_s > 0$: children differ not just in baseline ability, but in *how their trajectories accelerate* and in *when they accelerate*. No prior accumulator model frees these.
- **Input quantity is a small share of between-child variance.** $\pi_{\alpha} \approx 0.91$ across 5 samples (English long, Norwegian long, BabyView, SEEDLingS, Peekbank). Aligns with the meta-analytic $R^2 = 0.04\text{--}0.07$ of Coffey & Snedeker (2026).
- **LLMs are structurally different.** The same per-word sigmoid statistic gives $\kappa \sim 1$ for LMs vs. ~ 10 for children — a $10\times$ gap, with near-zero between-instance variance in LMs. Even Feng et al.'s CHILDES-trained models stay near the unit boundary.

Two signatures, twin constraints on theory

Calling back to the start

Acceleration: vocabulary grows super-linearly with age

Wordbank English (American) WS form. Each point = one child; lines = 10/25/50/75/90 percentiles via quantile regression



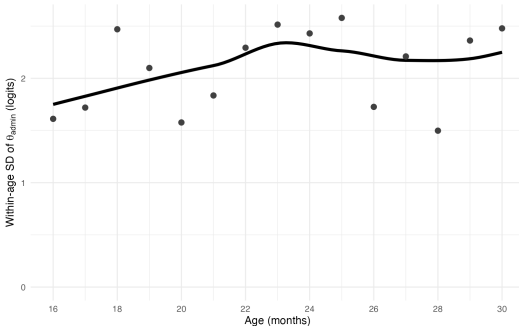
Acceleration.

$\kappa_{\text{pop}} \approx 10$ is above any version of pure accumulation.

Where does the super linearity come from?

Variability: within-age spread grows with age

Wordbank English (American) WS form. Per-admin theta from 1-PL Rasch IRT.



Variability.

$\pi_{\alpha} \approx 0.91$; input rate explains only $\sim 10\%$ of between-child variance.

Where does the other 90% live?